

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**Department of Computer Science and Engineering****ASSESSMENT TOOL 2 – Coding Challenge**

Course Code:	DSA0404	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 – Coding Challenge	Assessment:	Assessment Tool 2 – Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	B. Ganga Jayadev Reddy	Reg. No.:	192472248
Section / Batch:	CSE AI	Date:	25 - 08 - 2026

Code of Conduct

I B. Ganga Jayadev Reddy (192472248) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Ganga Jayadev Reddy

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In - d e p t h analysis with s t r o n g reasoning and	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Implement K-Means clustering from scratch (centroid initialization, assignment step, update step, convergence check) without using sklearn.cluster.KMeans. Handle the empty-cluster edge case (a centroid that gets zero points assigned).

Code :

```
import numpy as np

# Sample dataset
X = np.array([
    [1, 2], [2, 1], [1, 1],
    [8, 8], [9, 8], [8, 9],
    [5, 6], [6, 5], [5, 5]
])

k = 3
max_iter = 100

# Step 1: Initialize centroids randomly
np.random.seed(42)
indices = np.random.choice(len(X), k, replace=False)
centroids = X[indices]

for _ in range(max_iter):

    # Step 2: Assign each point to nearest centroid
    labels = []
    for point in X:
        distances = [np.linalg.norm(point - c) for c in centroids]
        labels.append(np.argmin(distances))
    labels = np.array(labels)

    # Step 3: Update centroids
    new_centroids = []
    for i in range(k):
        cluster = X[labels == i]

        # Empty cluster handling
```

```

    if len(cluster) == 0:
        new_centroids.append(X[np.random.randint(len(X))])
    else:
        new_centroids.append(cluster.mean(axis=0))

new_centroids = np.array(new_centroids)

# Step 4: Convergence check
if np.allclose(centroids, new_centroids):
    break

centroids = new_centroids

# Output
print("Final Centroids:")
print(centroids)

print("\nCluster Assignment:")
for i in range(len(X)):
    print(X[i], "-> Cluster", labels[I])

```

Output :

```

inal Centroids:
[[5.33333333 5.33333333] [1.33333333 1.33333333] [8.33333333
8.33333333]]
Cluster Assignment:
[1 2] -> Cluster 1
[2 1] -> Cluster 1
[1 1] -> Cluster 1
[8 8] -> Cluster 2
[9 8] -> Cluster 2
[8 9] -> Cluster 2
[5 6] -> Cluster 0
[6 5] -> Cluster 0
[5 5] -> Cluster 0

```




GPS Map Camera

Chennai, Tamil Nadu, India



Thandalam Sriperumbadur, Chennai, Tamil Nadu 602105, India

Lat 13.026439, Long 80.017213

Wednesday, 26/08/2026 11:14 AM GMT+05:30

Note : Captured by GPS Map Camera

